

John Yang

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Computer Science, Minor in Philosophy

May 2028

- **Courses:** Data Structures and Algorithms in C++, Object-Oriented Programming, Computer Organization, Discrete Math, Linear Algebra, Intro to Probability

EXPERIENCE

Software Engineering Intern

May 2026 – Aug 2026

Todd Agriscience

Los Angeles, CA

- Built a platform that runs crop growth simulations in parallel with **Dask** on NASA POWER and SoilGrids data, showing users how any soil-seed pairing performs against **5,000+** alternatives over 30 historical weather years.
- Built and optimized a web crawler that sources and ingests AI-relevant agricultural data from a range of news articles and blogs, lifting retrieval relevance and answer accuracy for the RAG-based knowledge system by **15%**.
- Refactored the entire **Supabase** authentication flow to consolidate scattered login, session, and token-handling logic, cutting auth-related code by **30%** and making the login system far easier to maintain, debug, and extend.

Undergraduate Researcher

Jan 2026 – Present

University of Michigan (Future of Programming Lab)

Ann Arbor, MI

- Extended a custom ReasonML DSL for **Hazel**, a live functional programming environment, so AI agents target code through AST definitions, bindings, and types instead of fragile line-range patches, cutting brittle edits by **40%**.
- Designed custom evaluation frameworks and ran systematic ablation studies across frontier models including Gemini and Claude, surfacing coding-agent tool-use quality across execution accuracy, composability, and trace legibility.
- Engineered structural insertion and multi-cursor bulk-editing capabilities for AI coding agents, cutting the discrete steps needed to apply complex, non-linear code changes and easing large, multi-part structural transformations.

Software Developer

Jan 2026 – Present

MaizeBus

Ann Arbor, MI

- Rebuilt the route and map screen in **Flutter** with interactive map components and redesigned navigation controls, sharpening real-time route visibility and everyday usability for a live student transit app with **1,500+** downloads.
- Debugged and resolved persistent navigation and routing issues across the app's screen flow while shipping new interface features, stabilizing app performance and smoothing the end-to-end user experience across core screens.
- Developed and maintained **RESTful API** endpoints in **Node.js** and **Express.js**, backed by MongoDB and deployed on **Heroku**, serving **300,000+** requests per day of real-time bus location and arrival-time data to riders.

PROJECTS

Auto On-Call Agent - AI Incident Response | *Python, FastAPI, REST API, Slack API*

- Built an autonomous incident-response pipeline that uses an LLM to pinpoint the commit responsible for an incident and automatically match alerts to the right response guide through a self-built RAG system, validated by successfully catching a **41x** error-rate spike across **62%** of traffic and auto-drafting a grounded incident report.

Nomble - Restaurant Finder | *React Native, TypeScript, Node.js, Express, PostgreSQL, Redis*

- Built a mobile app for Tinder-style restaurant discovery with swipeable cards and JWT-secured **REST** endpoints, using real-time location tracking to surface nearby spots via **Google Places API** and custom prompts to tailor recommendations, backed by a two-tier **Redis** cache that cut API calls and sped up per-user feeds.

Item Share - Splitting Costs With Roommates | *React, Next.js, TypeScript, Tailwind CSS, Supabase, SerpAPI*

- Made a roommate expense-splitting app where a **Next.js** backend calls **SerpAPI** to search and price-compare household items, stores results in Supabase, and runs a rules engine so roommates set a weekly spending cap or approval threshold, automatically splitting purchases and charging each card its exact share instantly through Stripe.

SKILLS

Languages: C/C++, Python, TypeScript, HTML/CSS/JavaScript, SQL, OCaml, Dart, R

Technologies: React, NextJS, React Native, Tailwind CSS, ThreeJS, NodeJS, ExpressJS, Redis, MongoDB, MySQL, Git, NPM, Docker, CI/CD, AWS, NumPy, Pandas, Dask, Scikit-learn, TensorFlow, Flutter, Bash, Linux, Figma

Interest: Avid Tennis & Pickleball Player, Aspiring Home Chef, Scenic Hikes, Photography, Recreational Runner